

Threat Intel Report on

Mitre Atlas Framework

BY

Team SafeNet

Mayur Pawar - 2033

Rahul Gouda - 2069

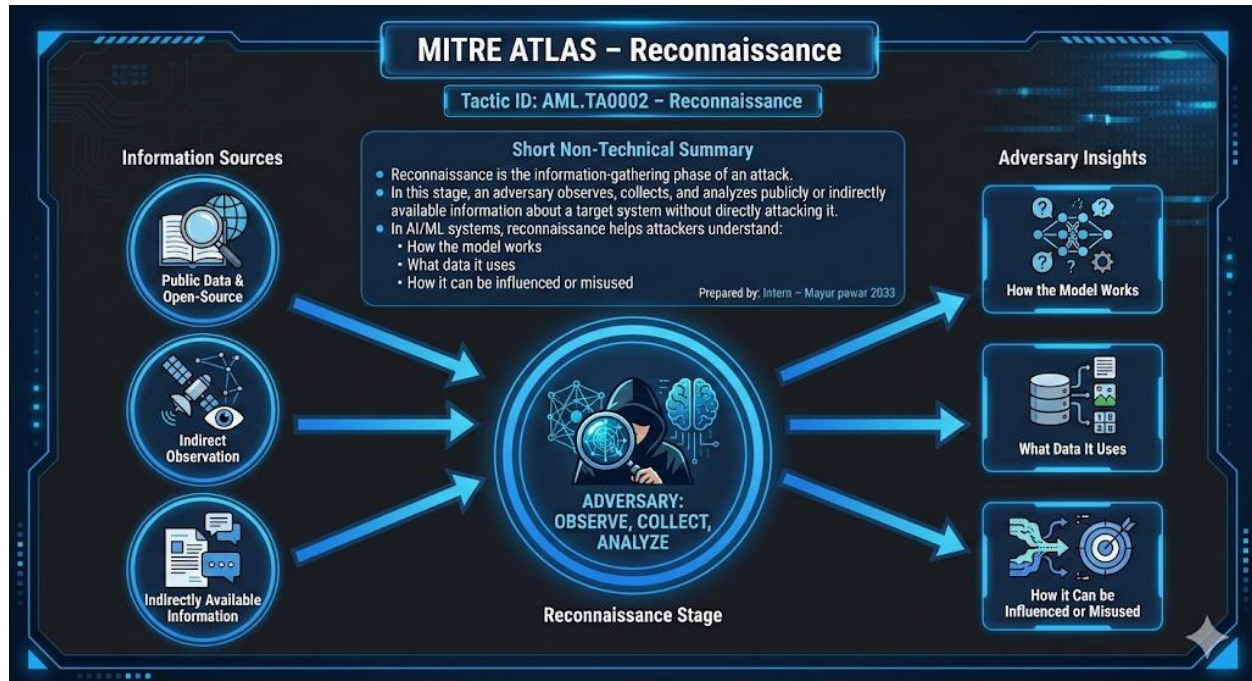
Janhavi Pandey – 2032

Sanika Jankar – 2038

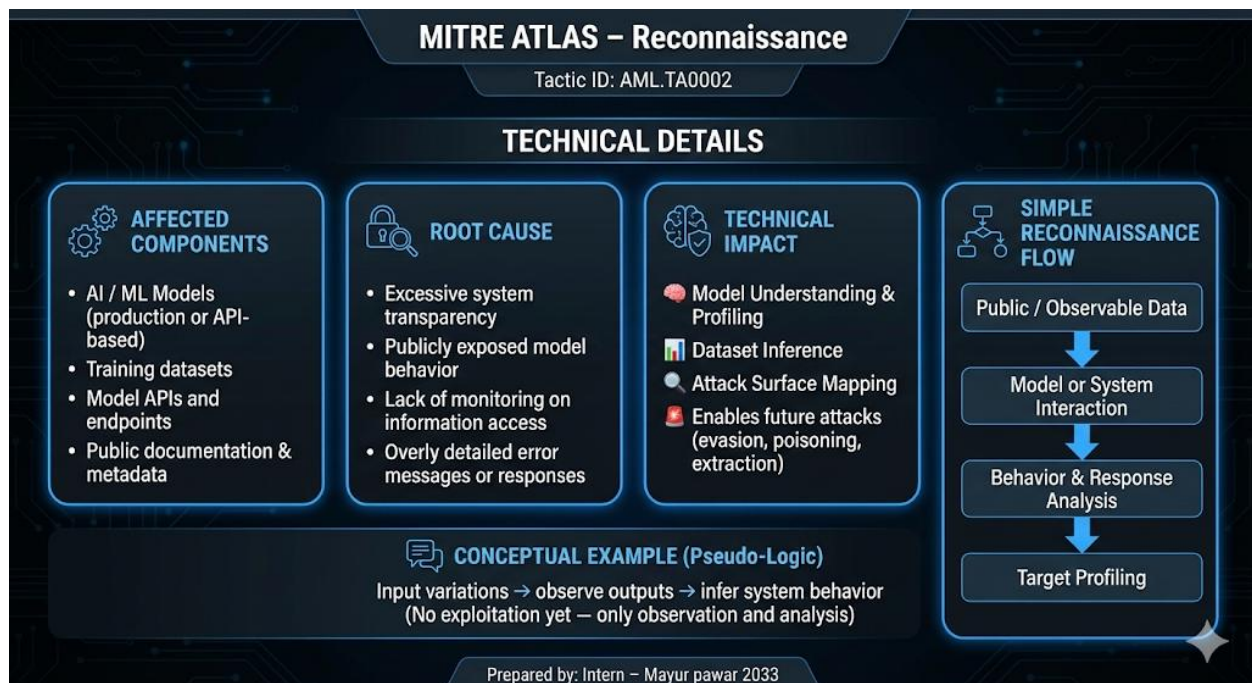
Shruti Rane - 2066

1. Reconnaissance

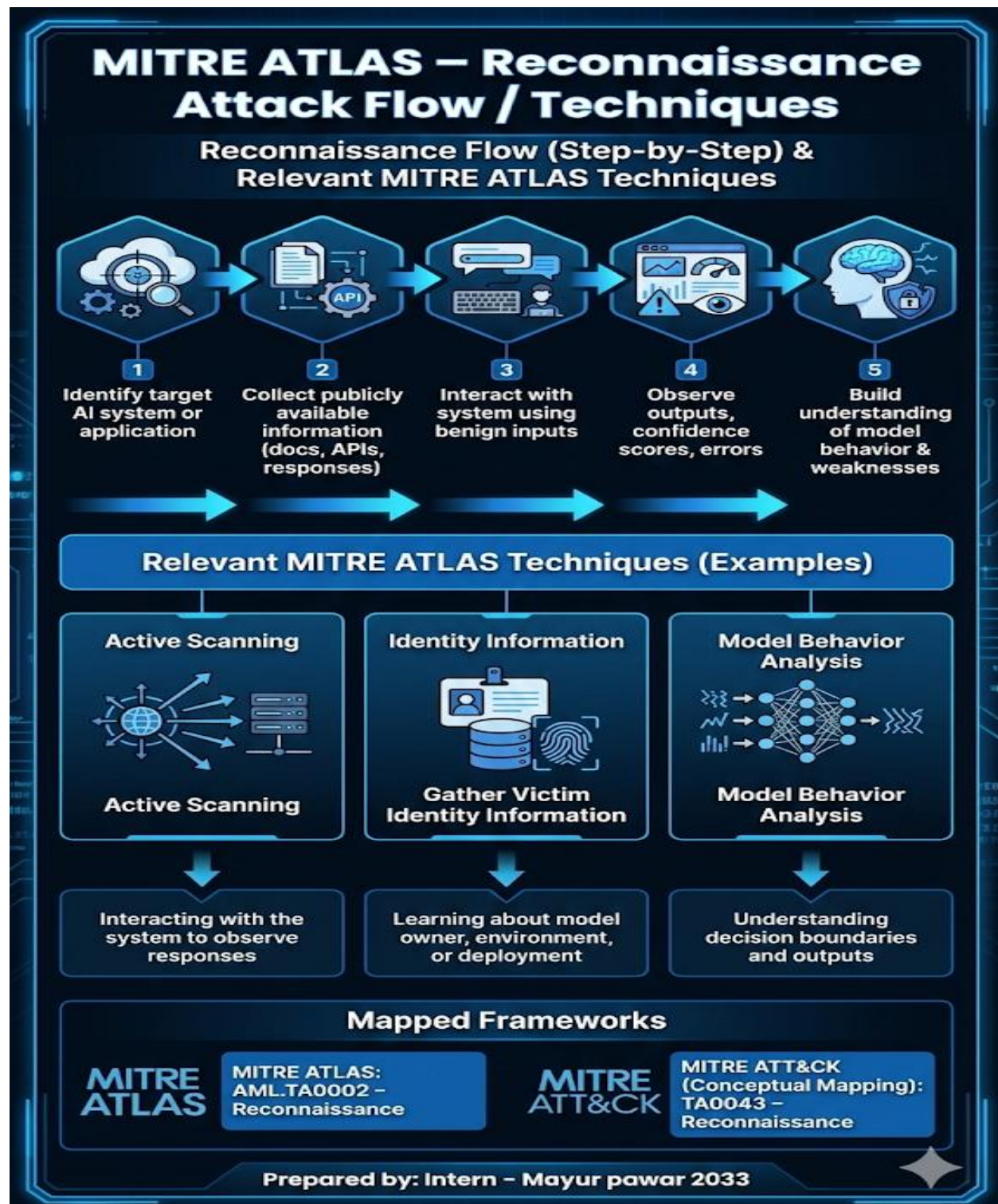
Overview



Technical Details:

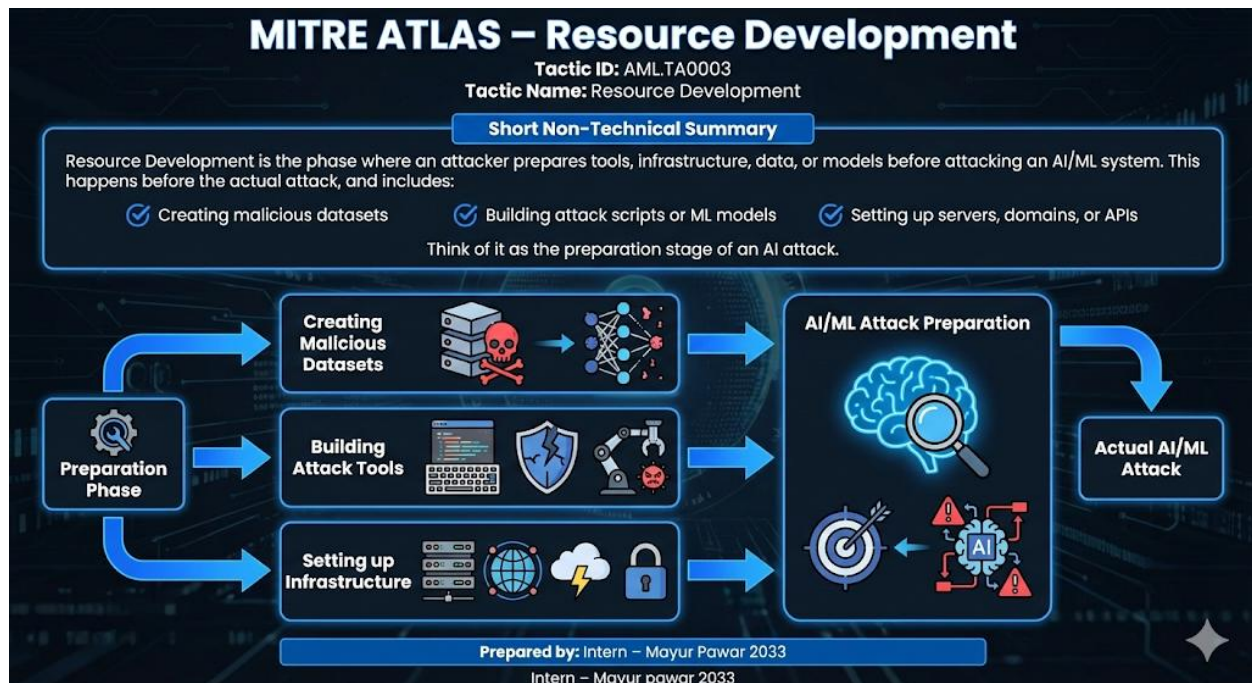


Attack Flow

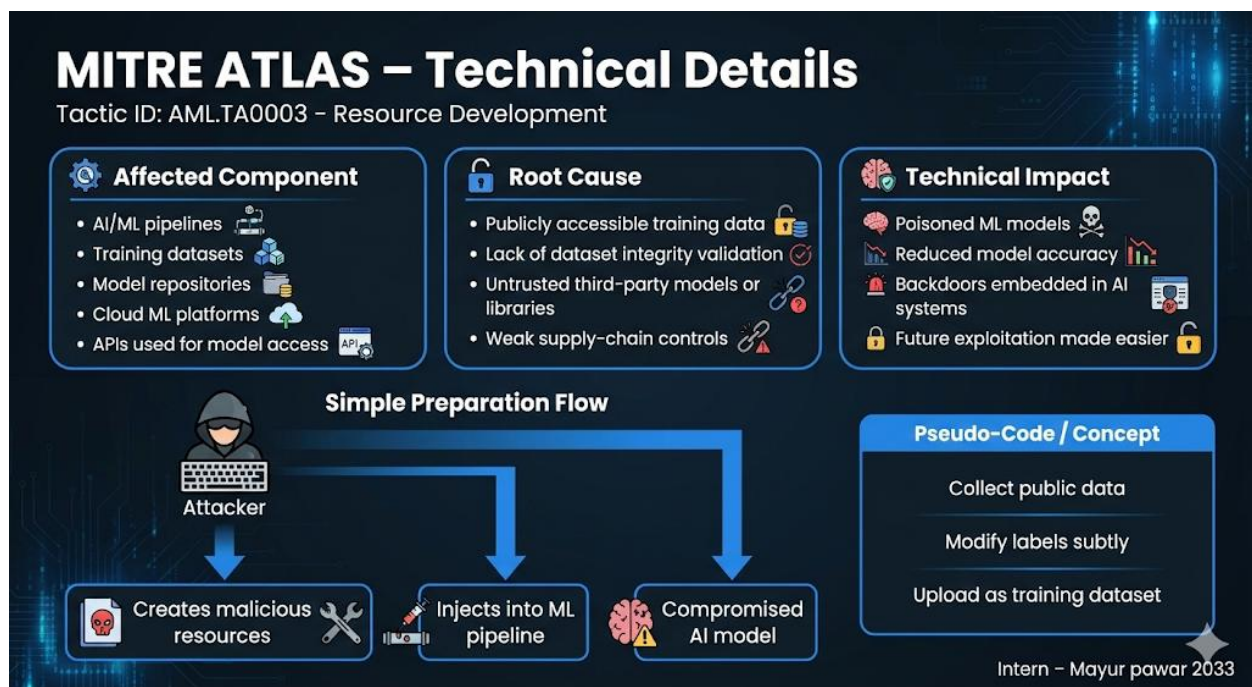


2. Resource Development

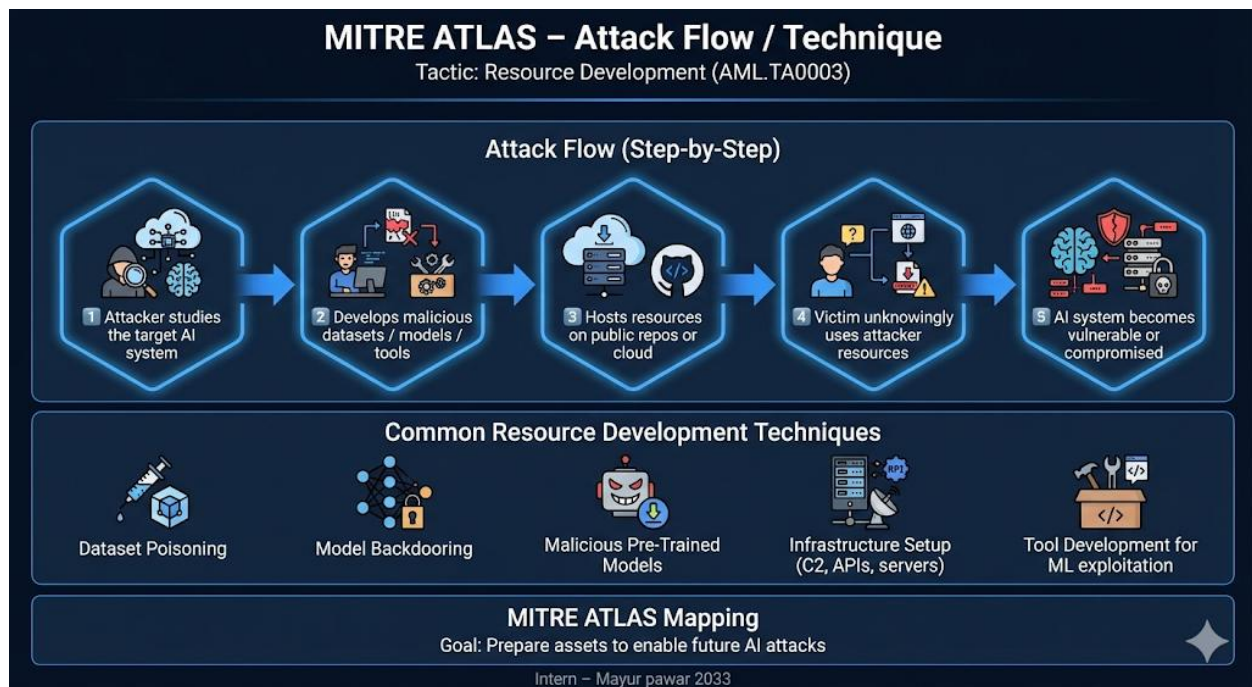
Overview



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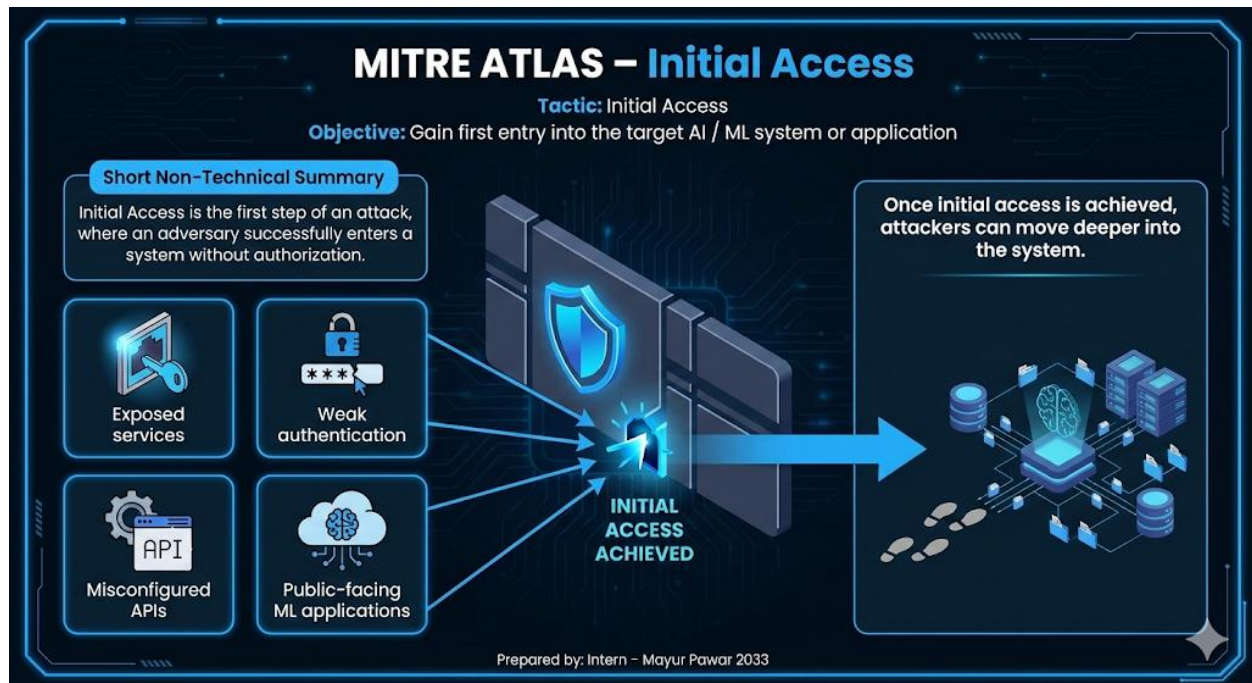


Attack Flow:

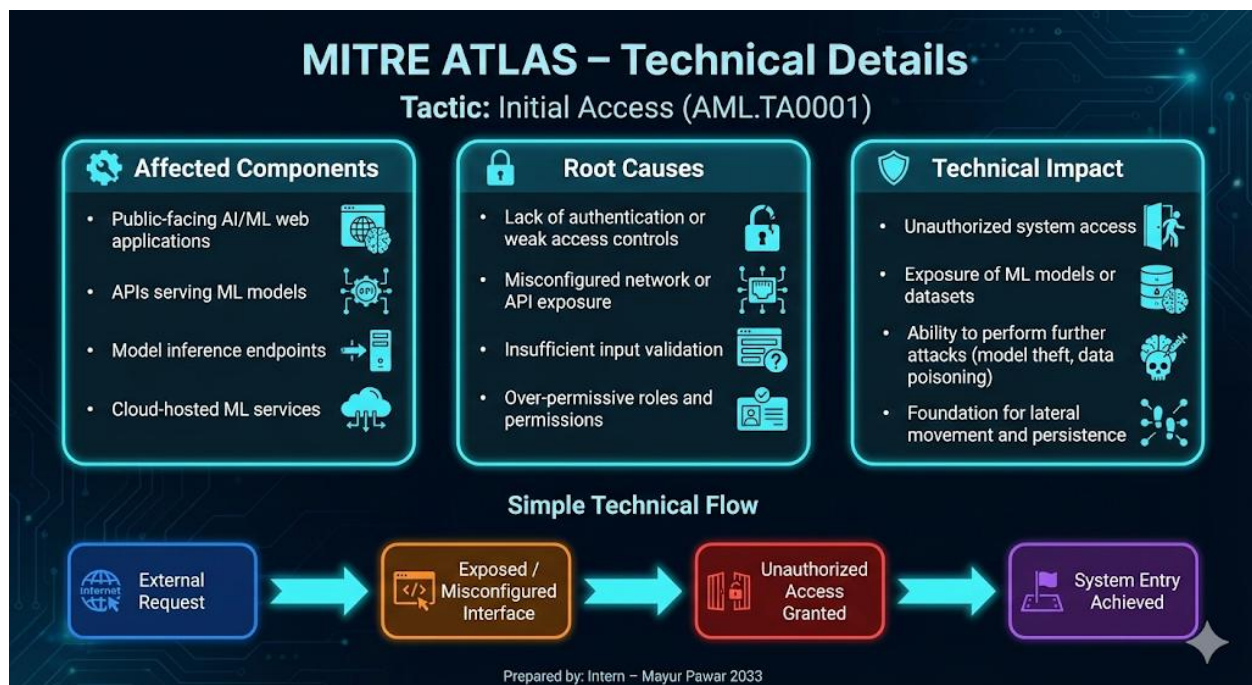


3. Initial Access

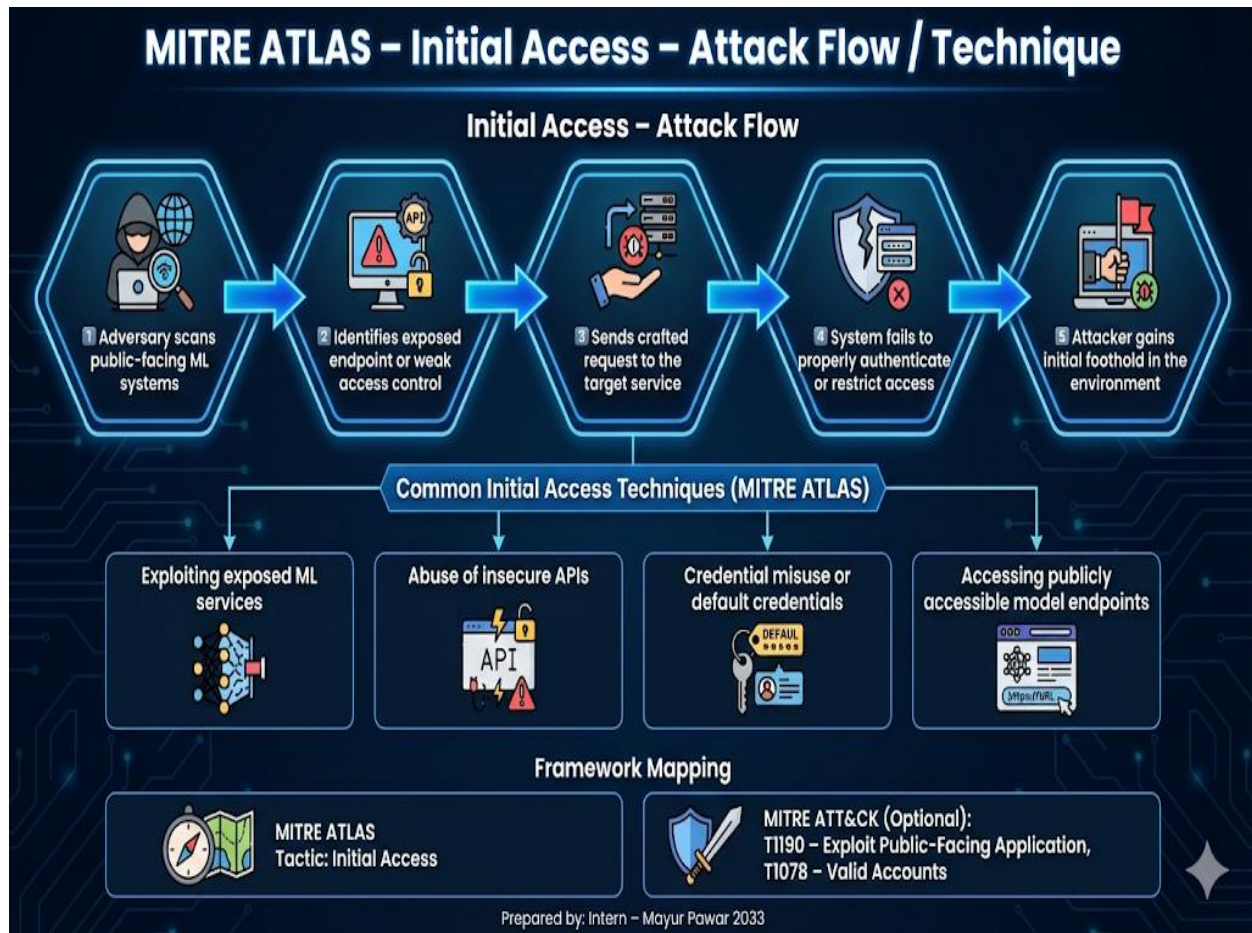
Overview



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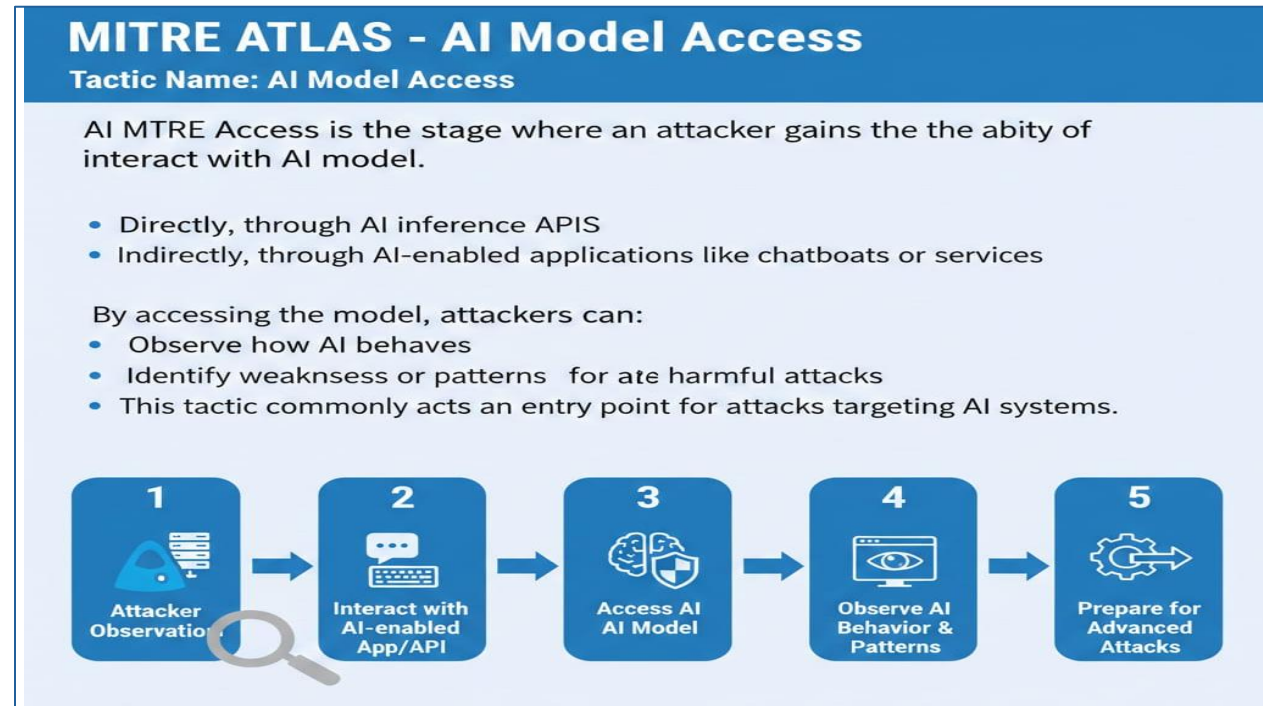


Attack Flow:

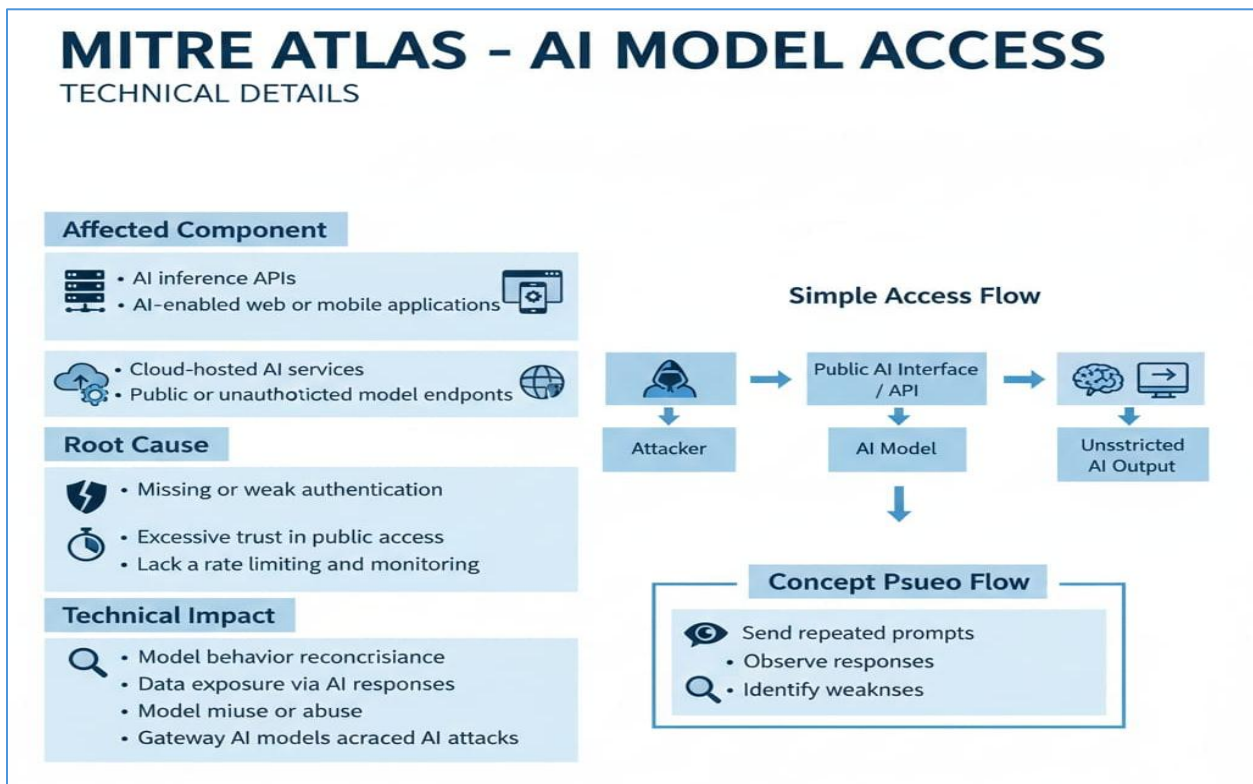


4. AI Model Access

Overview



Technical Details



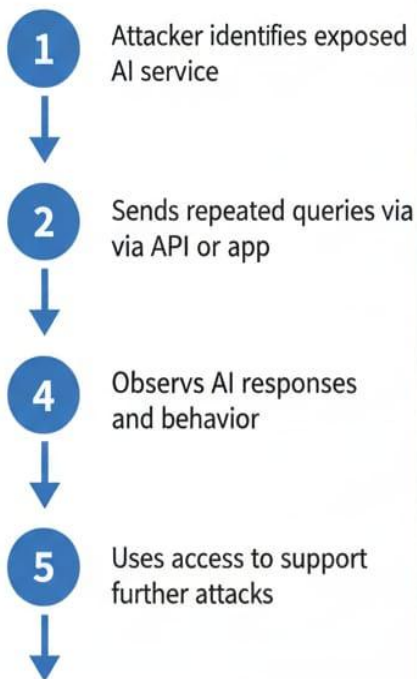
Attack Flow



MITRE ATLAS - AI MODEL ACCESS

ATTACK FLOW / TECHNIQUE

Attack Flow (Step-in-Step)



Common AI Model Access Access Techniques



AI Model Inference
API Access

Tactic: AI Model
Access (AML.TA0004)



AI-Enabled Product or
Service Abuse



MITRE ATLAS Mapping



Full AI Model
Access

Goal: Obtain interaction with AI
models to enable further
exploitation



Physical Environment Access
Access



5. Execution

Overview

MITRE ATLAS – Execution

Tactic Name: Execution

Tactic non-technical summary

Execution refers in the stage in an AI attack where adversary actions, of carry by this system. In this tactic, attackers make AI models, agents or AI-enabled software execute untincious instructions, such generiintend or malicious, runing commands, runing commands, or triggring automs auttimed actions. Executioms active miuse of AI capabilities and is progress towad data theft, system abuse, or data theft, sycabrise. This tactic. This tactic repreents where AI system actively participates in the attack.

Key Actions

-  Issue Malicious Prompts
-  Inject Malicious Data
-  Trigger Uniiitenited Commands
-  Automate Harmful Tasks

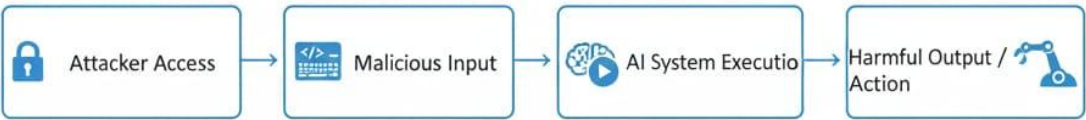
Technical Impact

-  Harmful Content Generation
-  System Command Execution
-  System Command Execution
-  Data Leakage
-  Service Disruption

MITRE ATLAS Mapping

-  Tactic: Execution (AML.TA0005)
-  Goal: Cause AI system to perform malicious actions

Execution Flow



Technical Details

MITRE ATLAS – Execution (Technical Details)

Affected Component

- AI Models
AI agents
- AI-enabled applications
OUS AI services
- Cloud AI services
- Prompt processing logic

Root Cause

- Input Valiation
- Over-privileed agents
- Exercution bundaries
- Blind trust
- Poor isolation

Technical Impact

- Malicious instructions
Unhahrized actions
- Harmful outputs
- Automatte attacher goals
- Data leakage

Root Cause

Simple Execution Flow

Attacker

Malicious Prompt / Instruction

</>

Agent

Unitended Execution

Malicious Outcome

Concept Flow

Crafting prompt

Bypasing safuguards

Blind trust

Triggens safuguards

Trigger execution

Obserivy harmful behavior

Attack Flow

MITRE ATLAS – Execution (Attack Flow)

Attack Flow (Step-by-Step)



Attacker gains access to an AI system or application



Attacker crafts a malicious prompt or instruction



Prompt bypasses AI safety controls or guardrails



AI model or agent executes unintended actions



Malicious output or action is produced

MITRE ATLAS Mapping



Tactic: Execution (AML.TA0005)



Goal: Cause AI systems to perform untended or malicious actions

Common Execution Techniques



- LLM Prompt Injection

- Guardrail Bypass



- Tool or Plugin Abuse

- AI Agent Command Execution



- Indirect Prompt Injection

6. Persistence

Overview

MITRE ATLAS – Persistence

Tactic Name: Persistence

- Persistence refers to the stage where the attacker continues to influence control over an AI system for a long period of time.
- In this tactic, the attacker modifies the AI configurations, data sources, or system behavior so that malicious or harmful effects remain active even after restarts, updates, or patches, or normal use.
- Persistence allows attackers to repeatedly exploit or manipulate AI systems without access each time. This tactic focuses on maintaining long-term presence within AI-enabled environments.

Key Actions



Modify AI Configurations



Backdoor Training Data



Hijack Automation Pipelines



Schedule Malicious Tasks

Technical Impact



Undetected Manipulation



Recurring Malicious Outputs



Persistent Access



Long-Term System Compromise

MITRE ATLAS Mapping



Tactic: Persistence (AML.TA0006)



Goal: Maintain long-term control over AI systems

Persistence Flow



Attacker Access



Malicious Configuration / Data



AI System Modification



Persistent Control



Technical Details

MITRE ATLAS – Persistence (Attack Flow)

Attack Flow (Step-h-Step)

1.  Attacker gains initial access to an AI system
↓
2.  Attacker modifies AI agent configuration or system prompt
↓
3.  Attacker injects poisoned data into RAG or knowledge sources
↓
4.  AI system stores the malicious configuration or data
↓
5.  AI system stores the malicious configuration or data
↓
5.  Malicious behavior persists across sessions, restarts, or updates
↓

Common Persistence Techniques

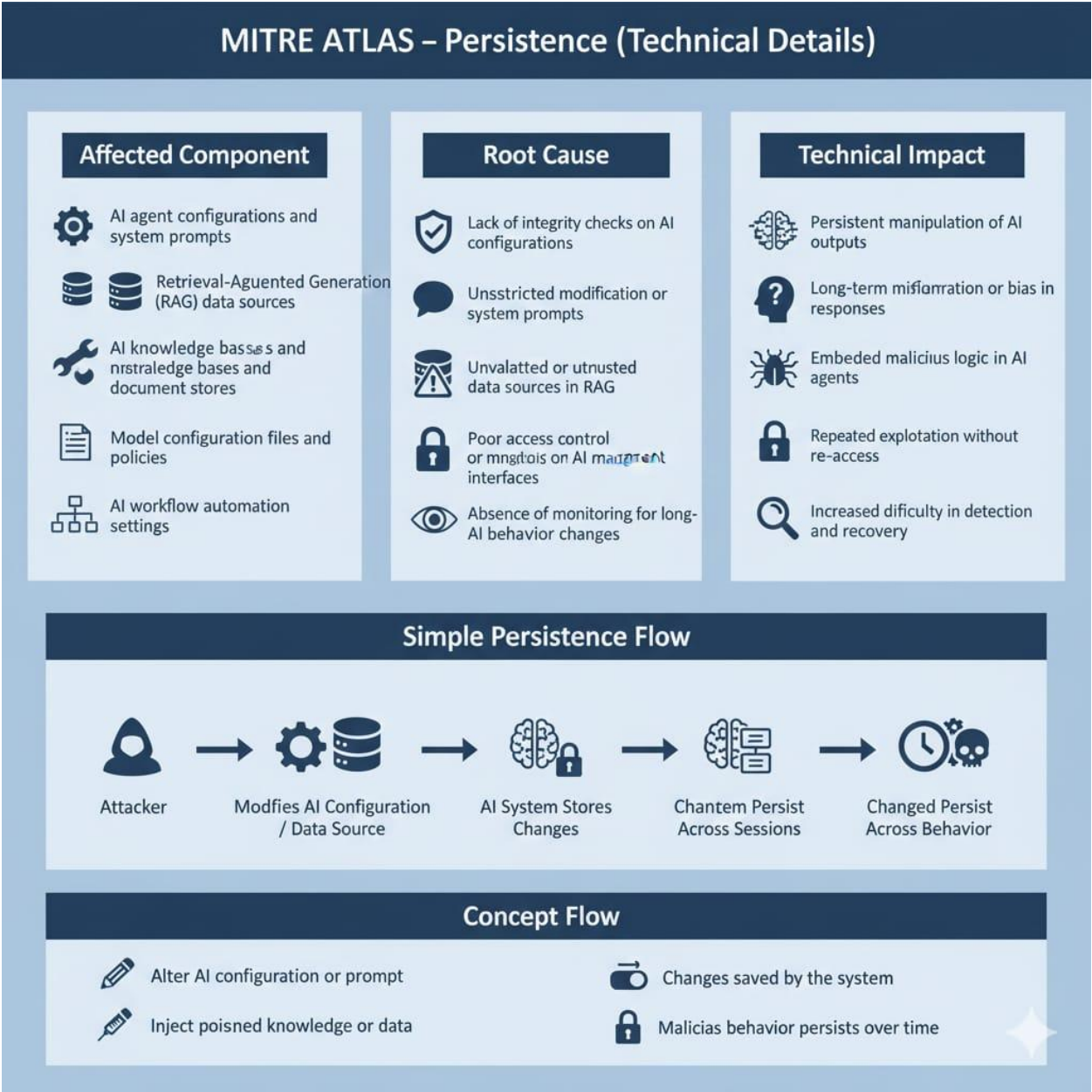
-  RAG Poisoning
-  Modify AI Agent Configuration
-  RAG Poising
- Persistent System Prompt
-  Knowledge Base Tampering
-  Long-Term AI Behavior Manipulation

MITRE ATLAS Mapping

-  Tactic: Persistence (AML.TA0006)
Goal: Maintain long-term influence or control over AI systems

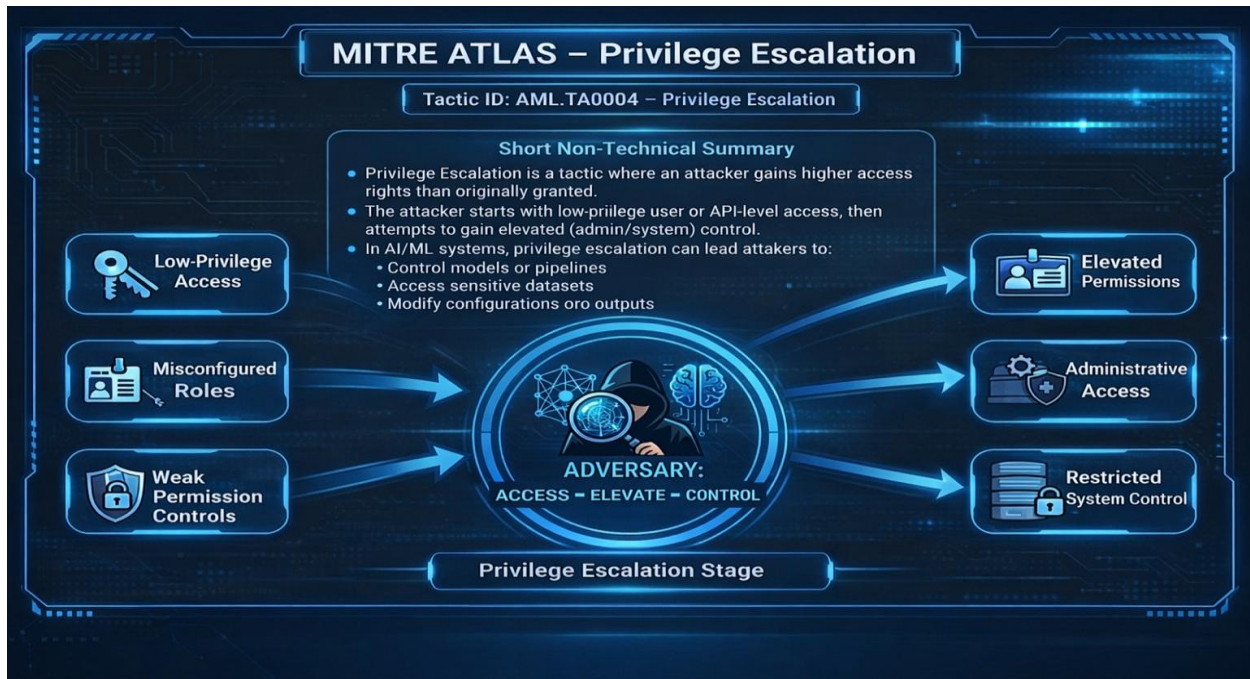


Attack Flow

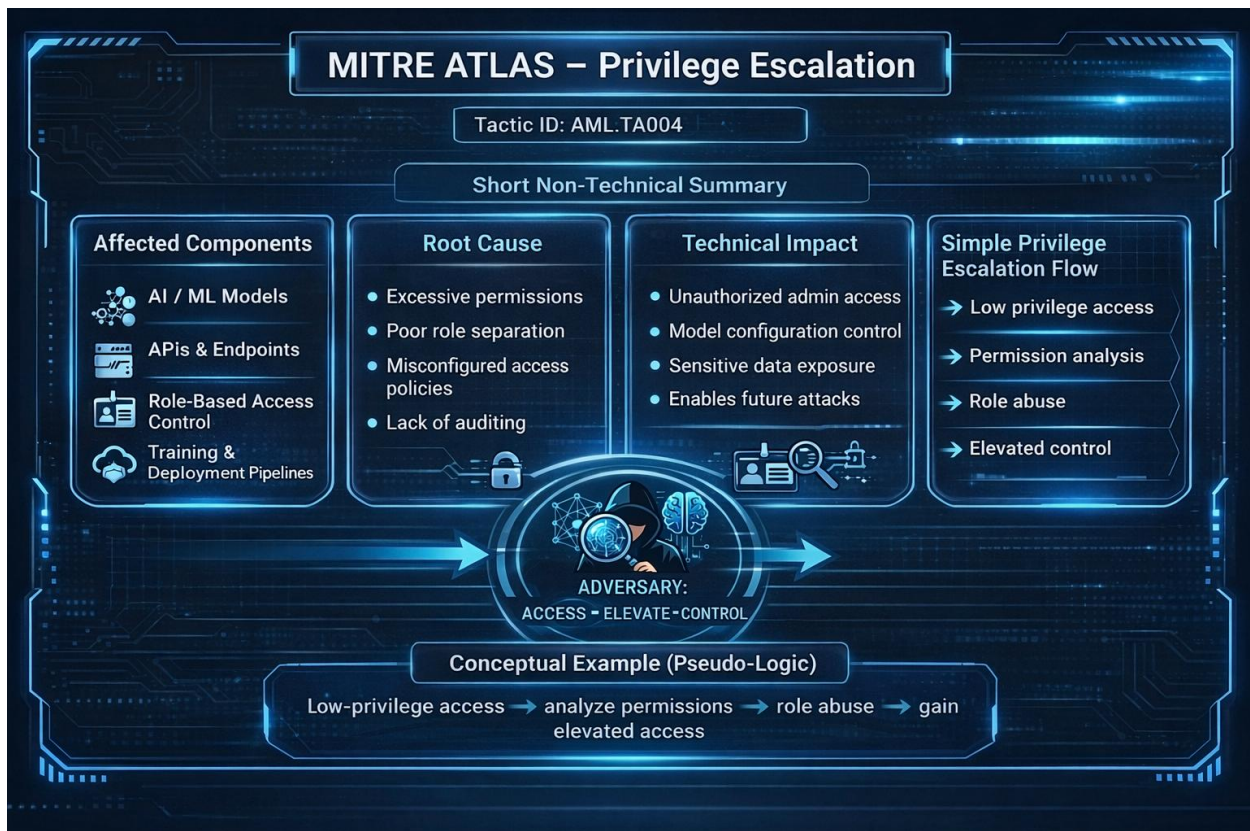


7. Privilege Escalation

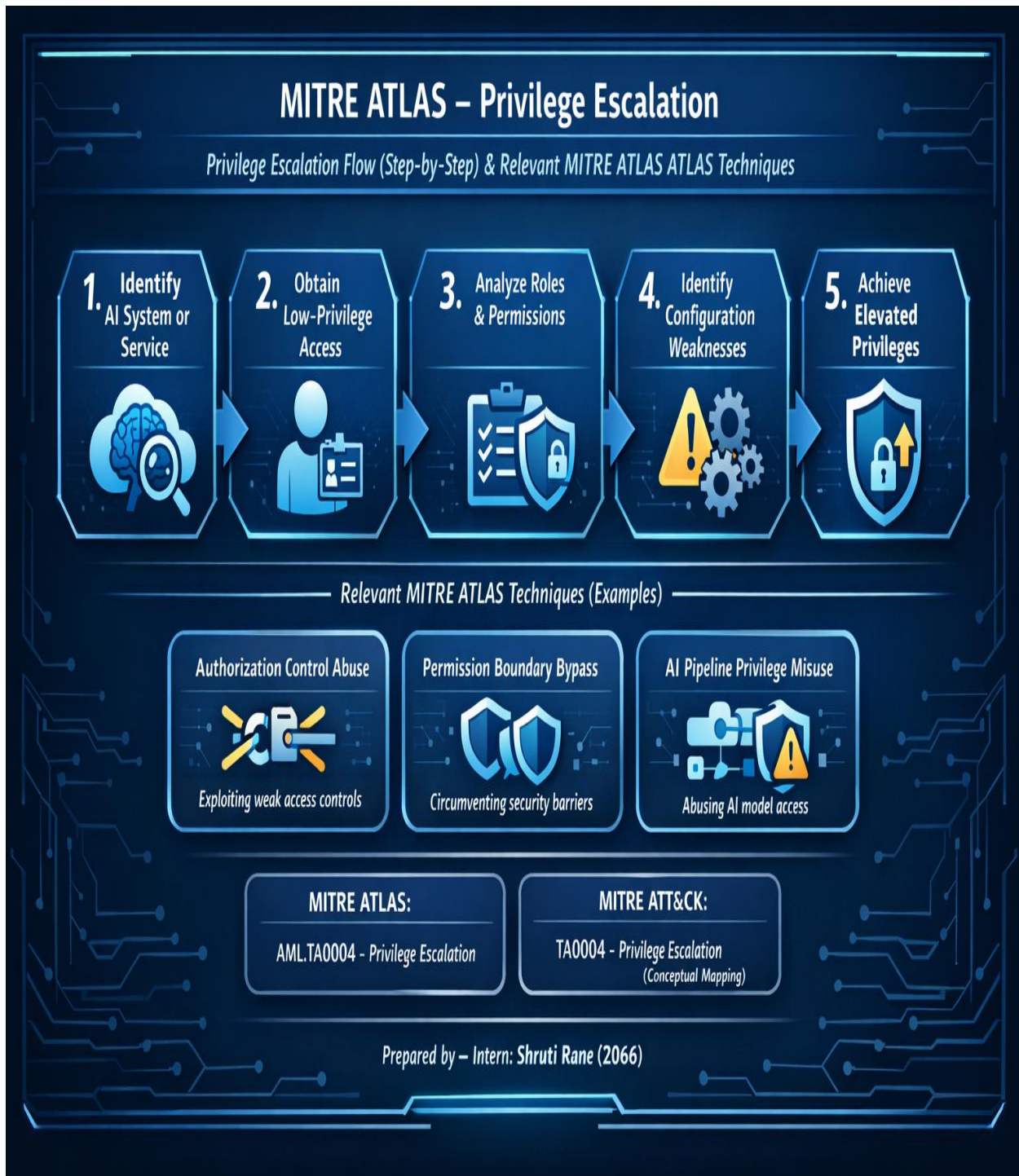
Overview



Technical Details



Attack Flow

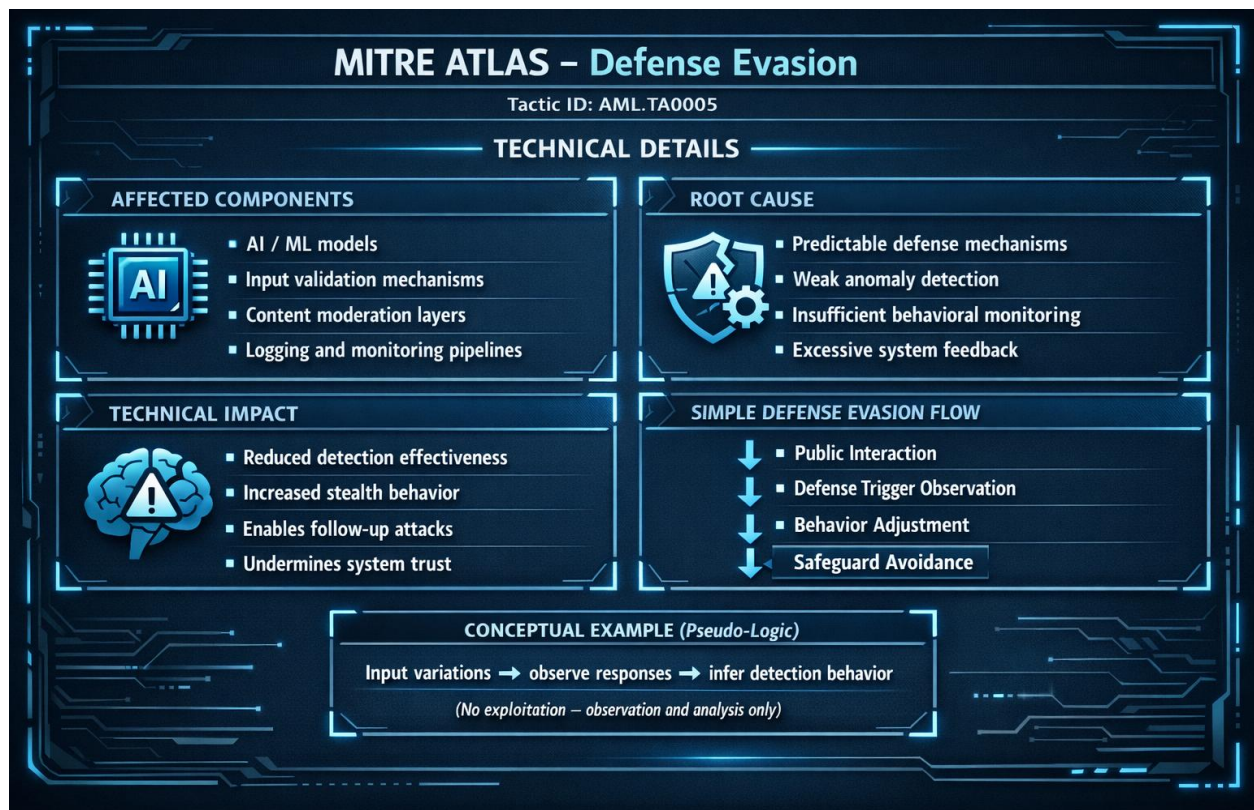


8. Defense Evasion

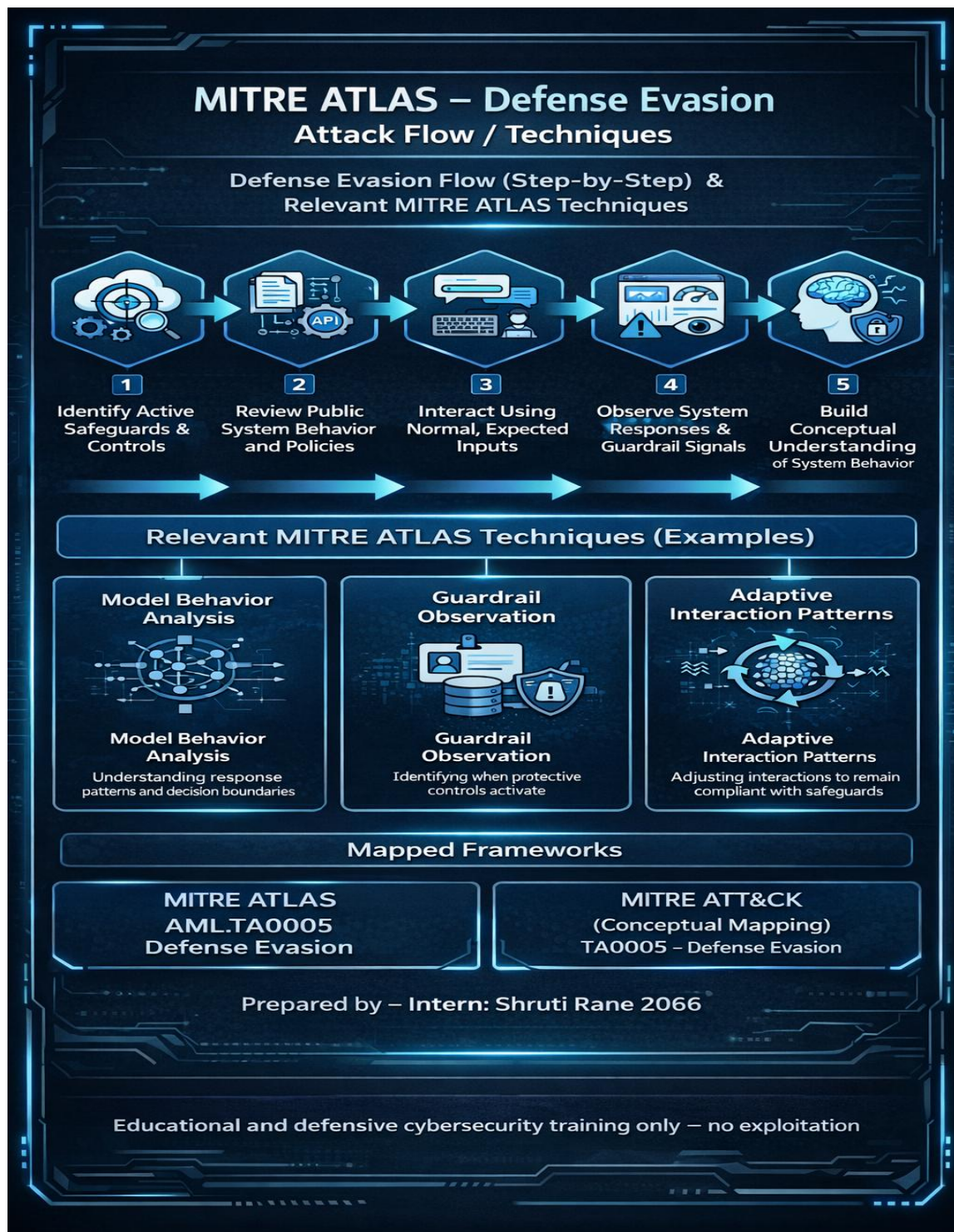
Overview



Technical Details:

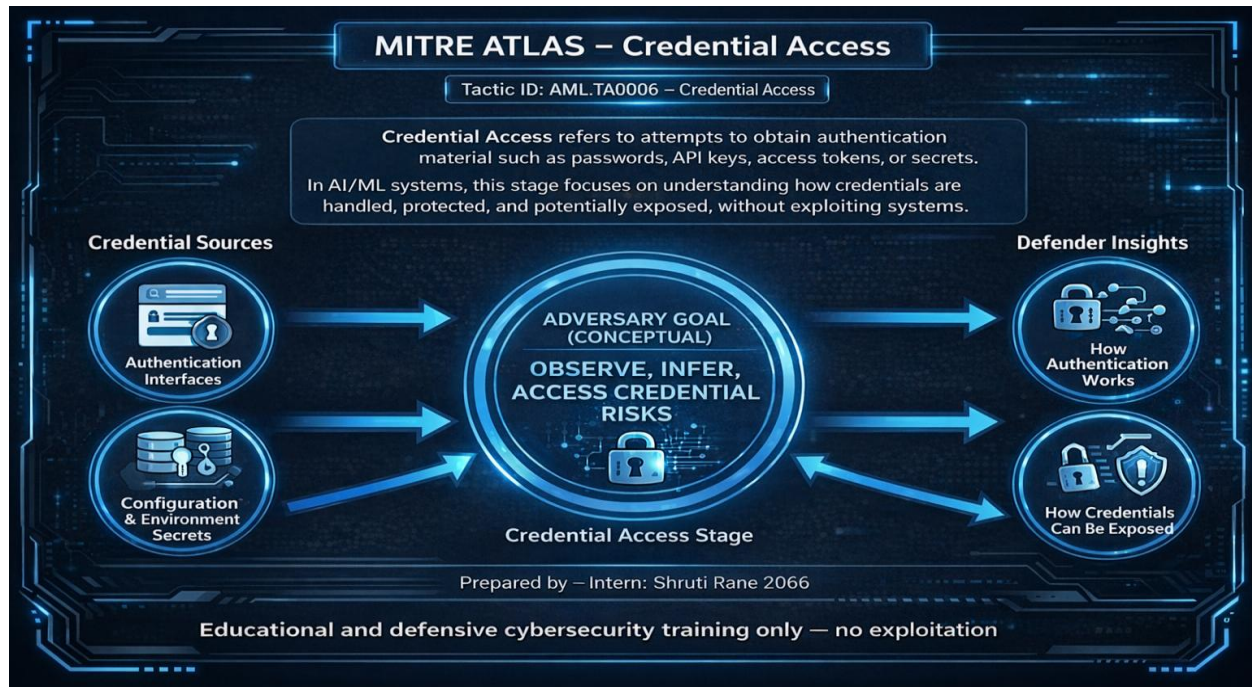


Attack Flow:

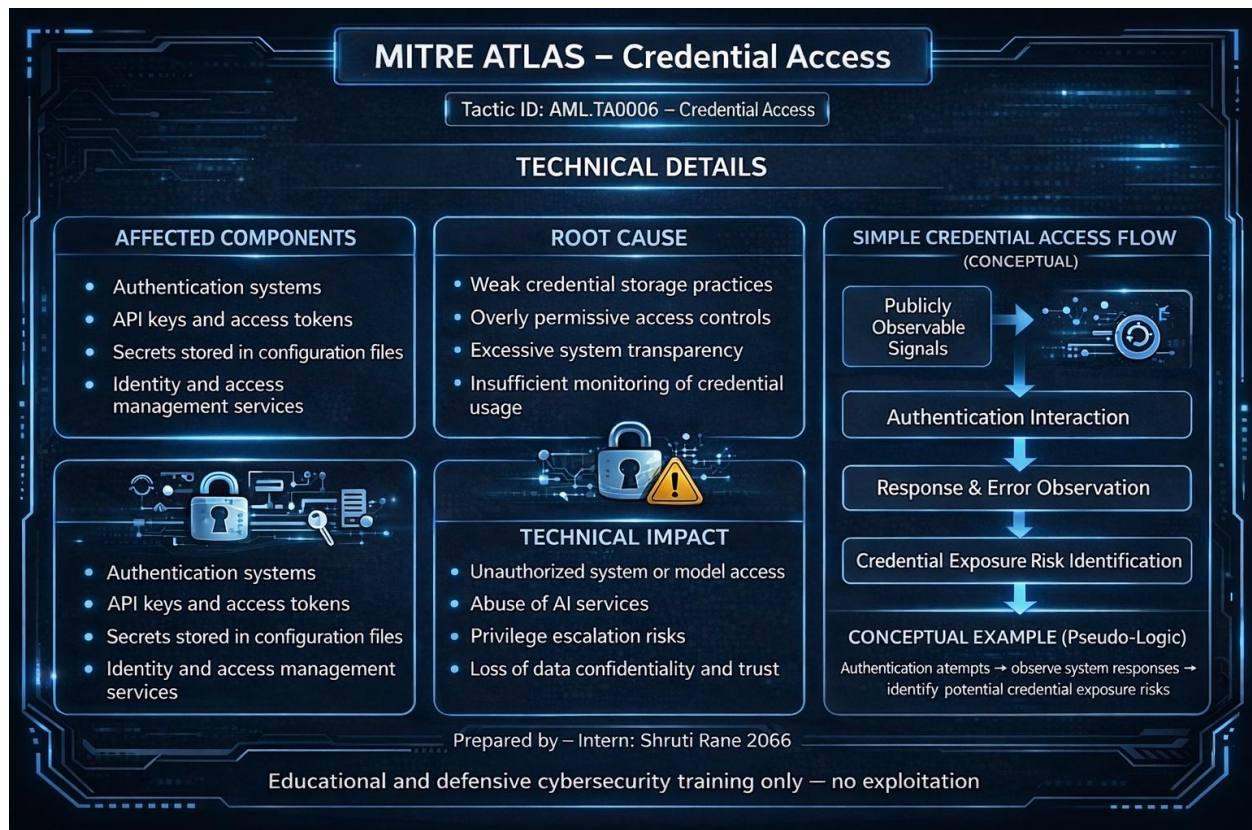


9. Credential Access

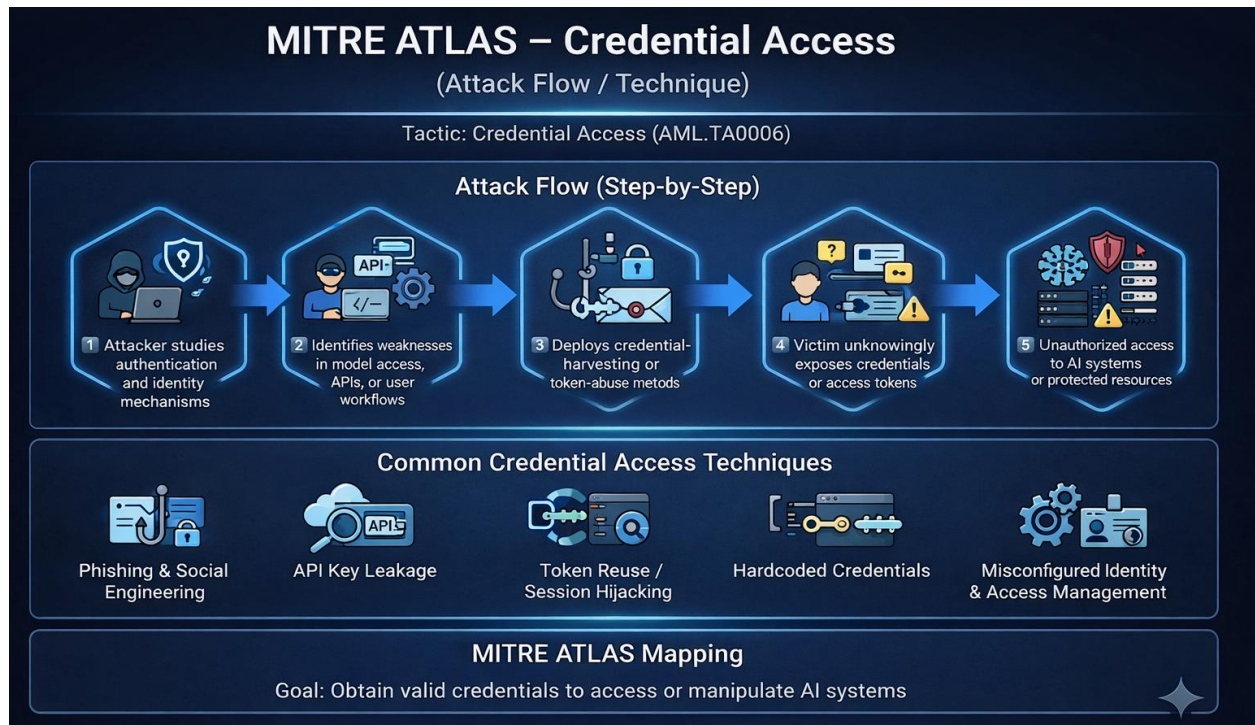
Overview



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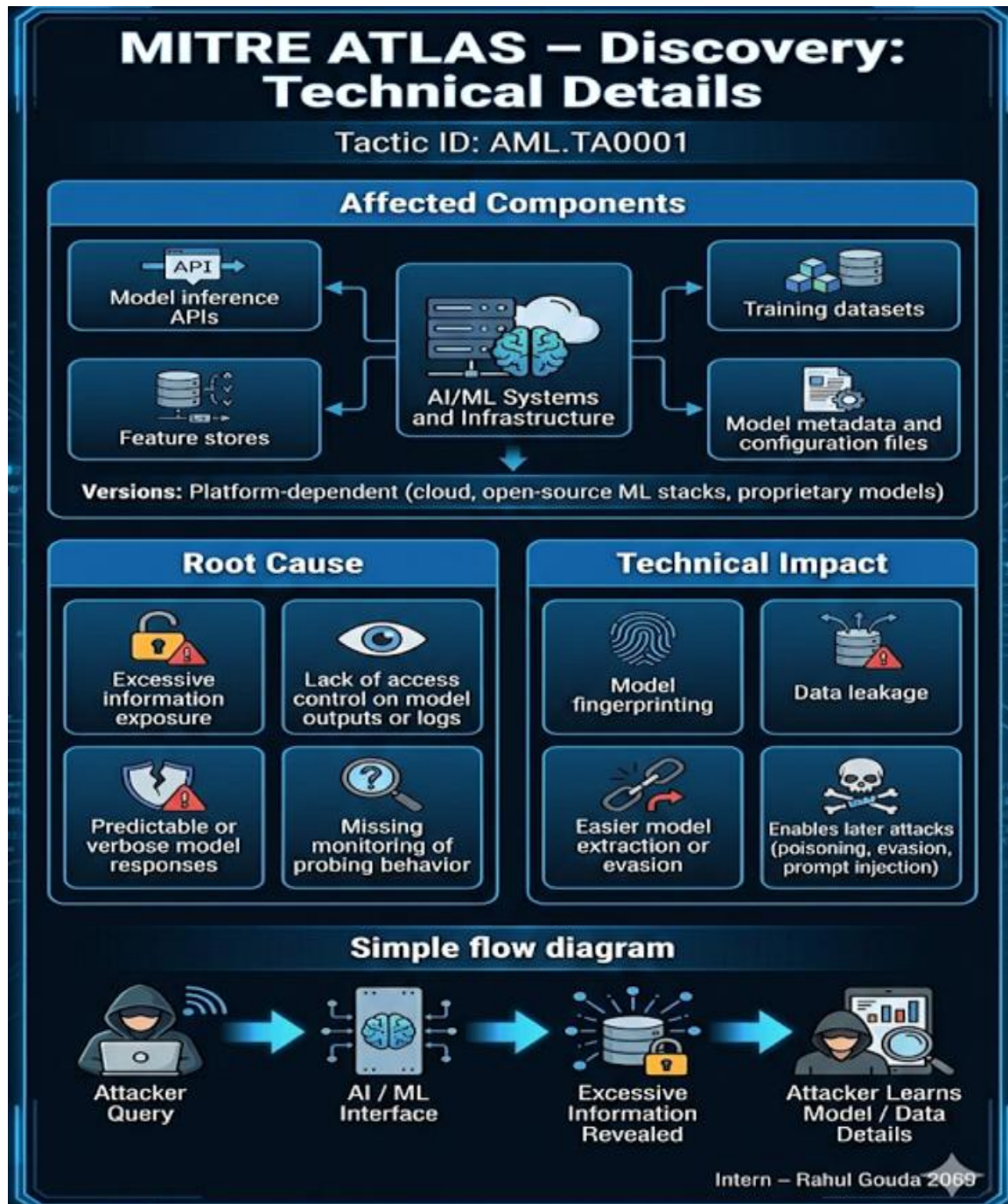


Attack Flow

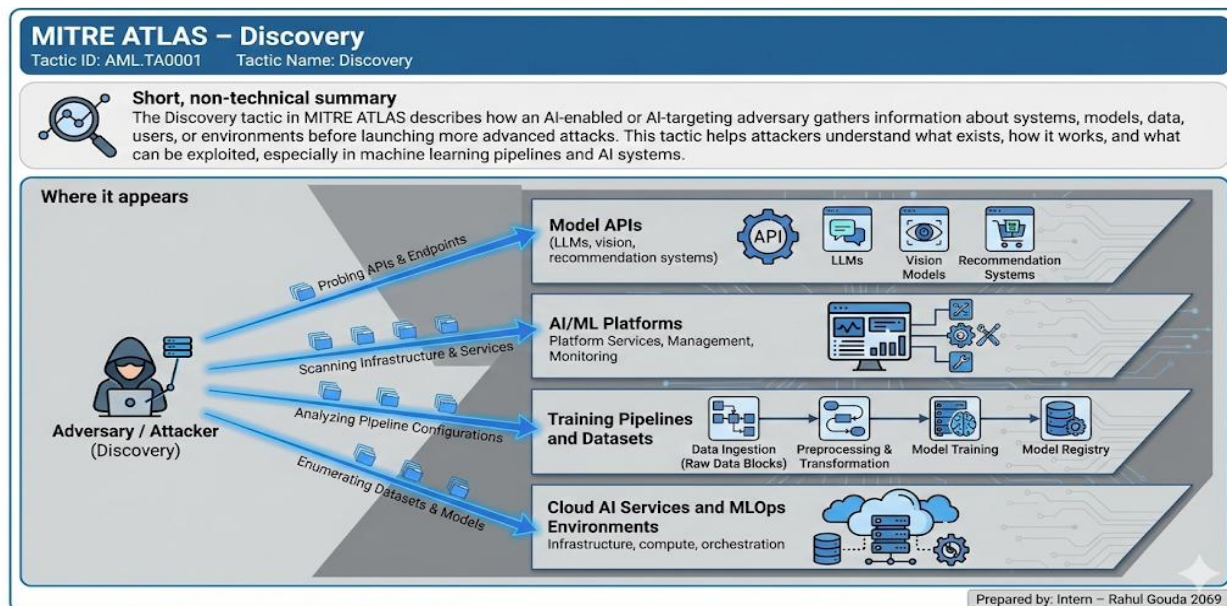


10. Discovery

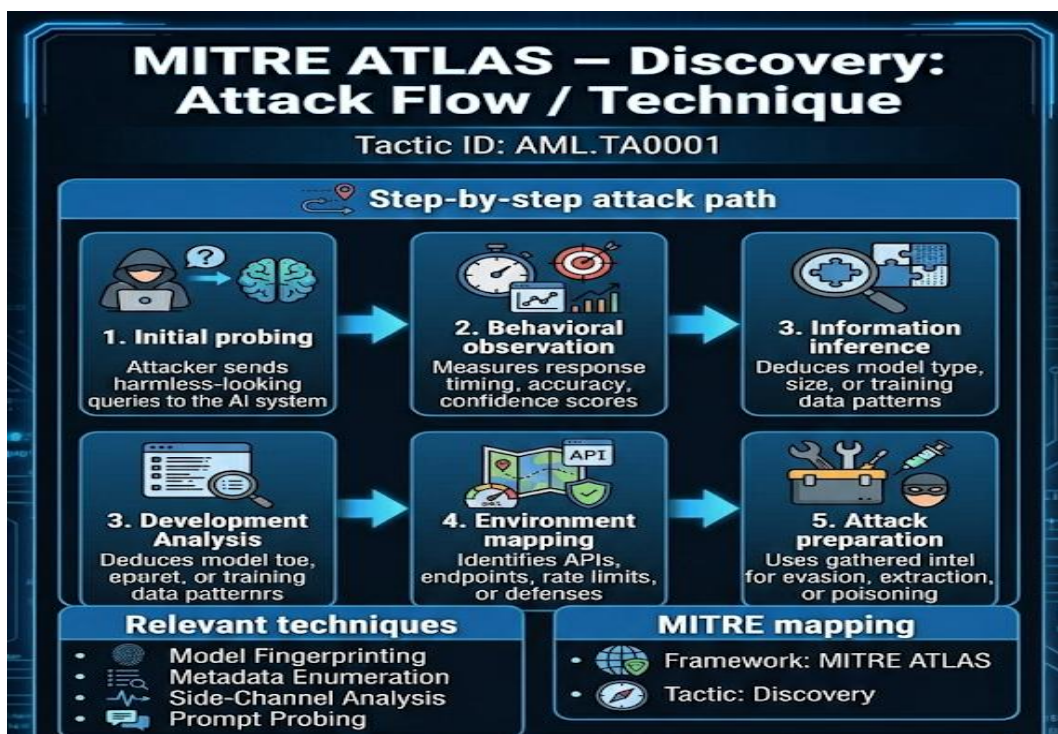
Overview



Technical Details

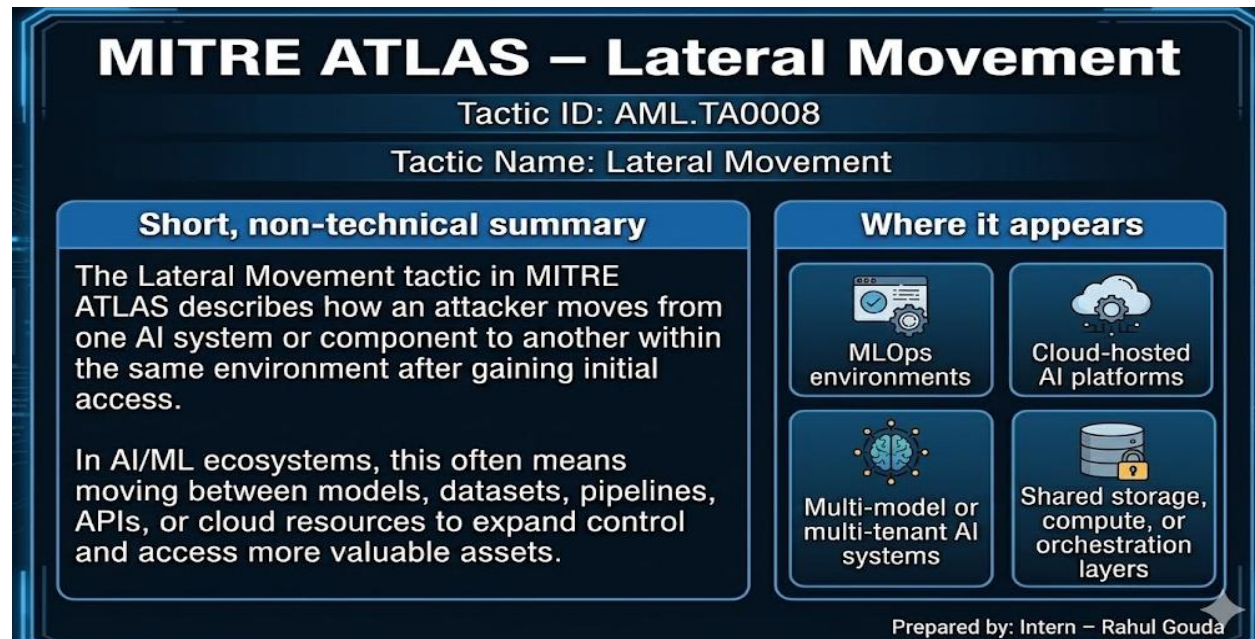


Attack Flow:

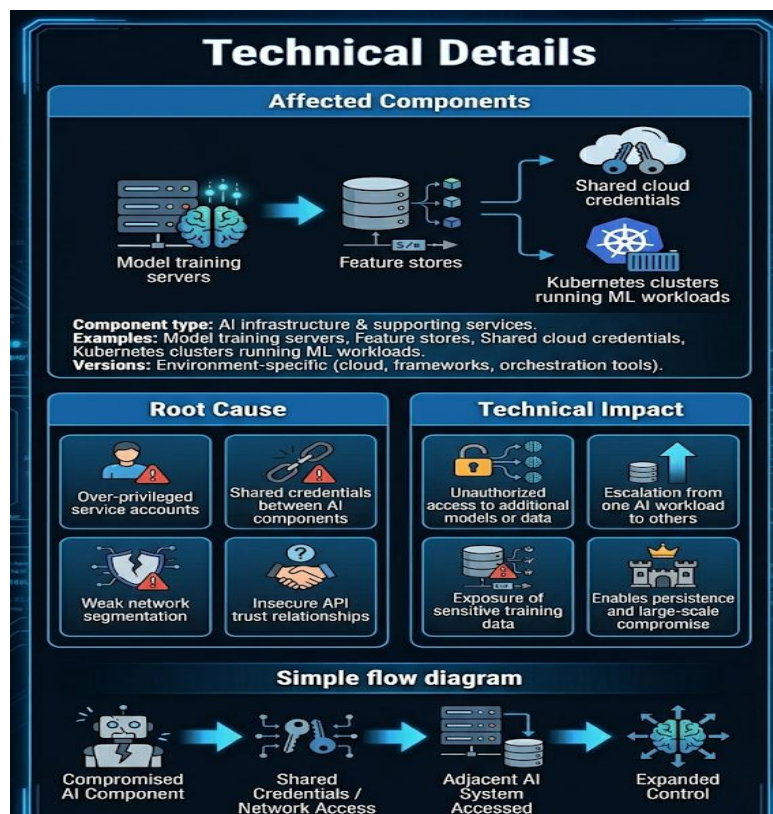


11. Lateral Movement

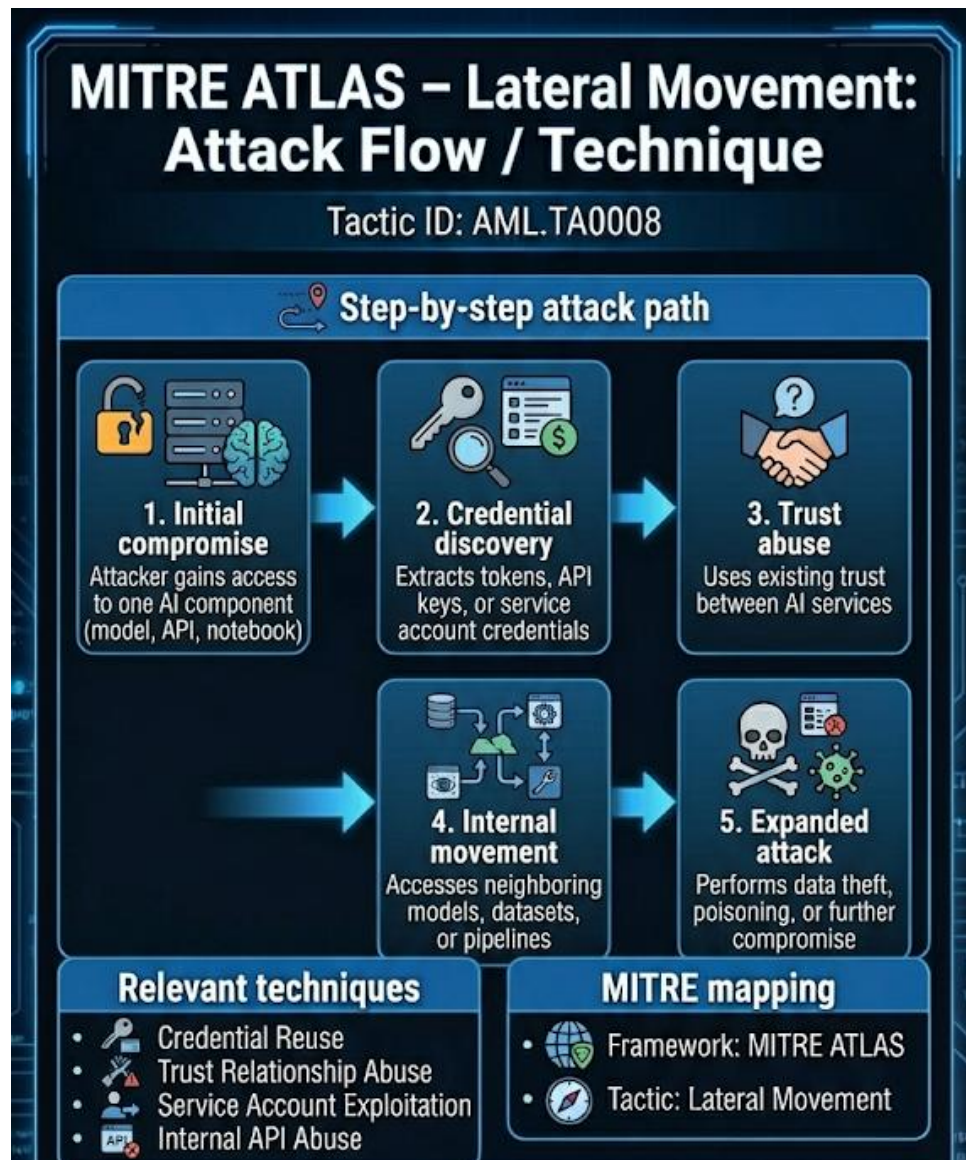
Overview



Technical Details:



Attack Flow:



12. Collection

Overview

MITRE ATLAS – Collection

Tactic ID: AML.TA0005

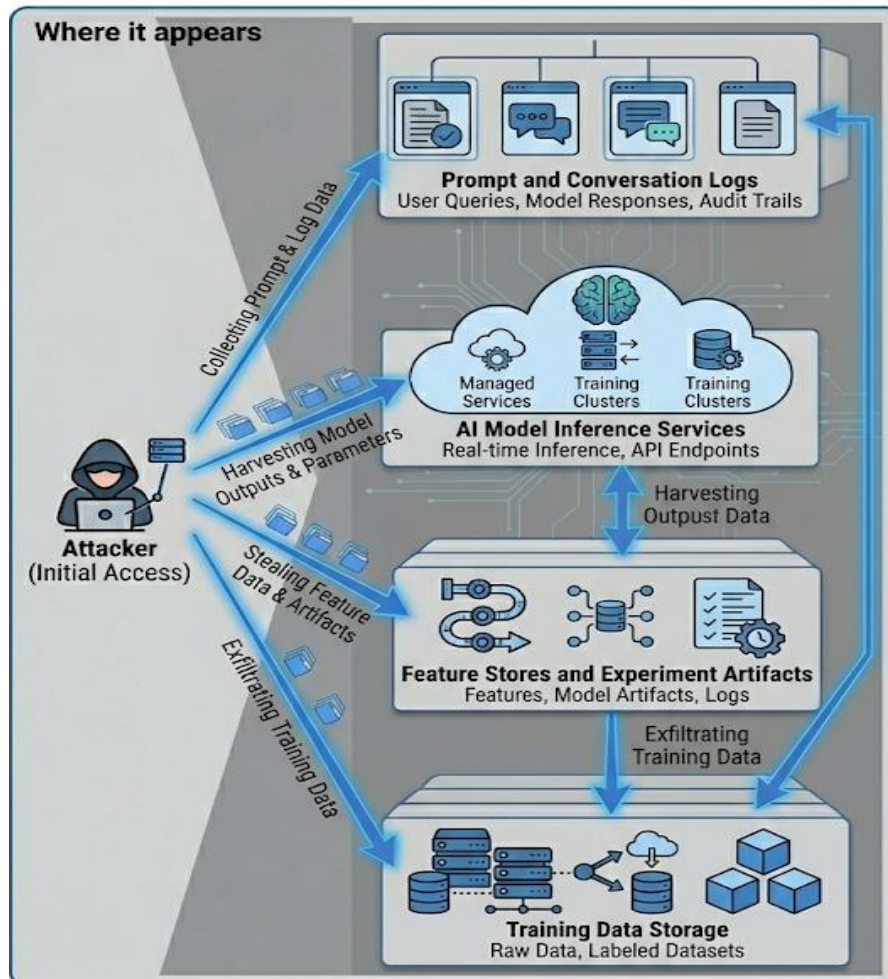
Tactic Name: Collection



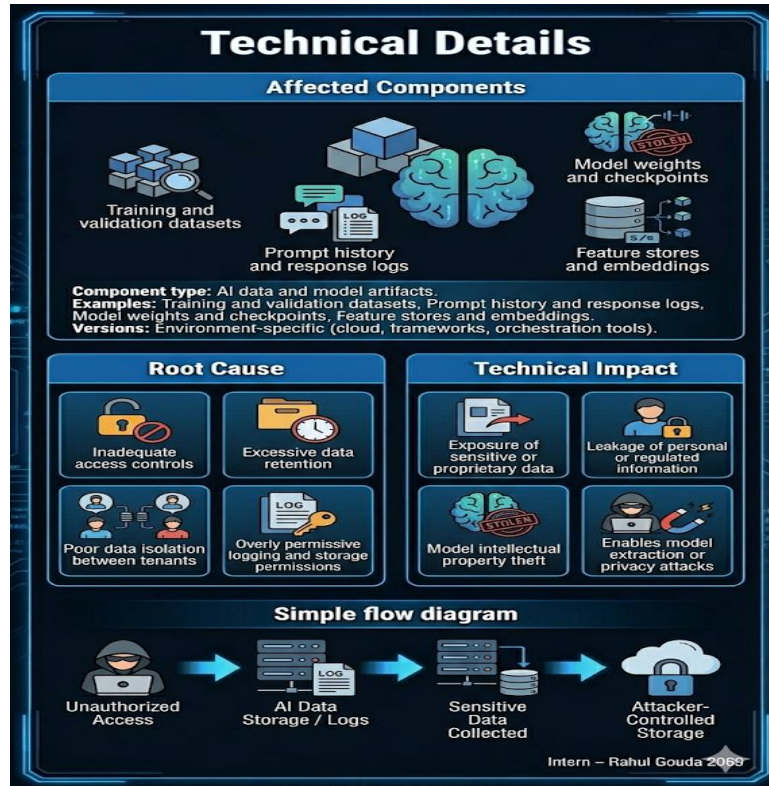
Short, non-technical summary

The Collection tactic in MITRE ATLAS describes how an attacker gathers valuable data from AI systems after gaining access. This data may include training datasets, model outputs, prompts, logs, or model parameters, which can later be stolen, analyzed, or used to improve future attacks.

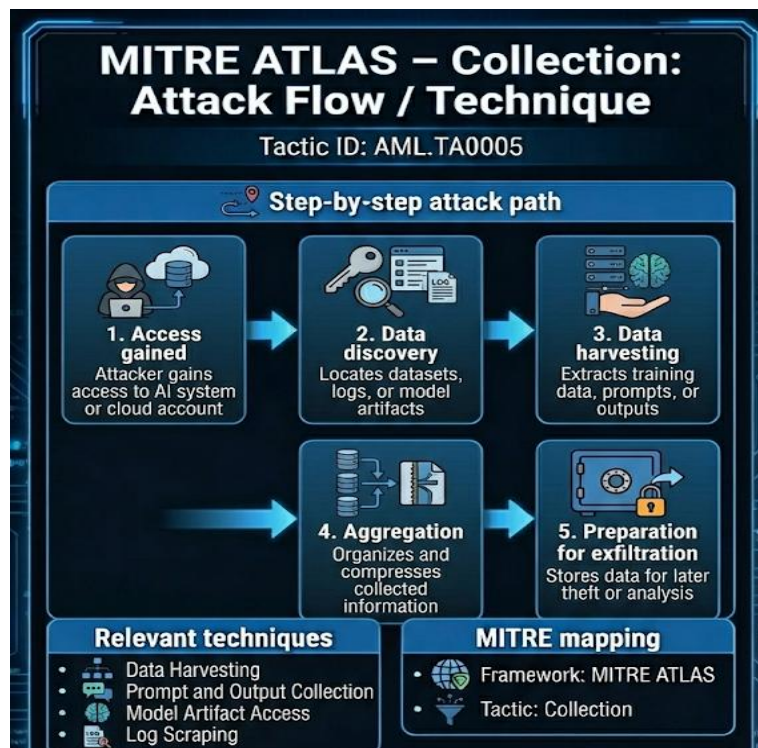
Where it appears



Technical Details:



Attack Flow:



13. Command and Control

Overview

MITRE ATLAS

COMMAND AND CONTROL

Tactic ID: AML.TA0011

Tactic Name: Command and control

Short Non-Technical Summary

Command and control is the phase where an attacker prepares tools, infrastructure, data, or models before attacking an AI/ML system.

This happens **before the actual attack**, and includes:

- Creating malicious datasets
- Building attack scripts or ML models
- Setting up servers, domains, or APIs

Think of it as the **preparation stage** of an AI attack.



• **Creating malicious datasets**



Building attack scripts or ML models



Server & Domain Setup

Think of it as the **preparation stage** of an AI attack.

Prepared by: Intern – Sanika Jankar

Technical detail:



Attack Flow:



14. Exfiltration

Overview

MITRE ATLAS

Exfiltration

Tactic ID: AML.TA0010

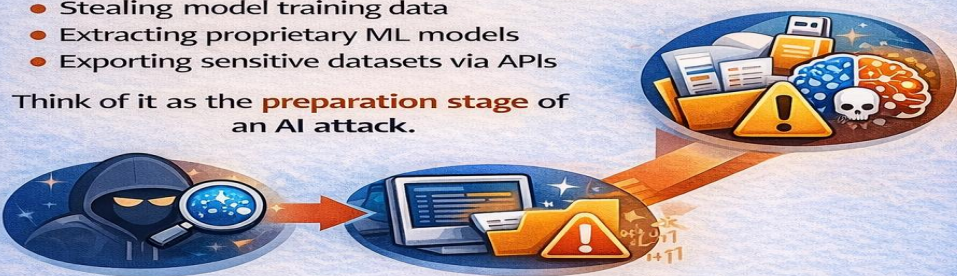
Tactic Name: Exfiltration

Short Non-Technical Summary

Exfiltration is the phase where an attacker prepares tools, infrastructure, data, or models before attacking an AI/ML system. This happens **before the actual attack**, and includes:

- Stealing model training data
- Extracting proprietary ML models
- Exporting sensitive datasets via APIs

Think of it as the **preparation stage** of an AI attack.



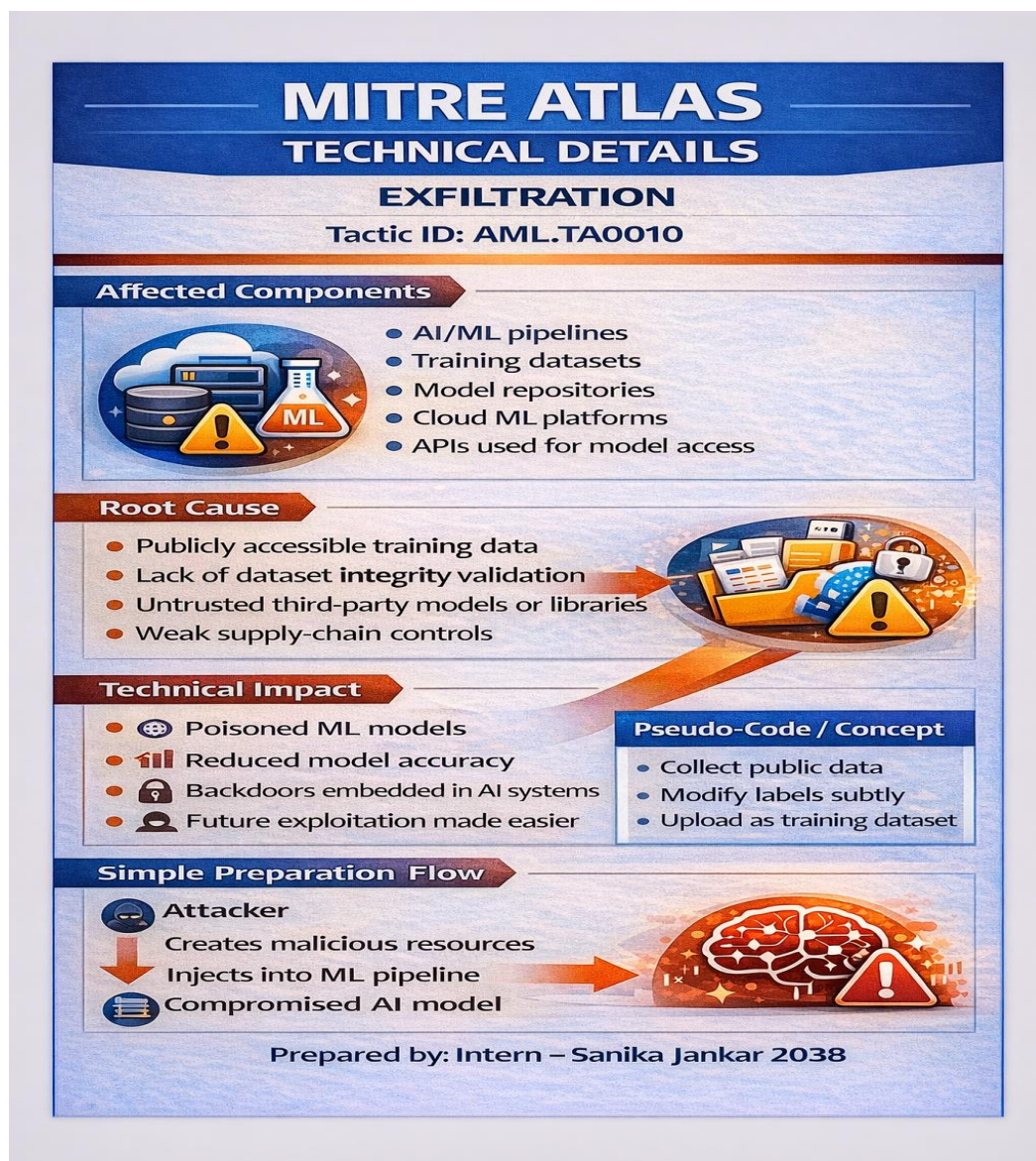
Think of it as the **preparation stage** of an AI attack.

Q: Start Non-Technical Summary

Exfiltration - Stealing model training data 	- Model Backdooring - Exporting sensitive datasets via APIs
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Prepared by: Sanika Jankar 2038

Technical detail:



Attack flow:



15. Impact

Overview

MITRE ATLAS – Impact

Tactic ID: AML.TA0040

Short Non-Technical Summary
the attacker phase where attacker tools toools infrastucture data or before attacking AI/ML system.

This happens before actual attack, and includes:

- Creating malicious datasets
- Building attack scripts or ML models
- Building attack scripts or ML models
- Infrass-tucture domains, or APIS



Prepared By: Intern – Sanika Jankar 2038

Think of ais as preparation stage of an AI attack.



Technical detail:

MITRE ATLAS – Impact

Tactic ID: AML.TA0040

Affected Component

AI/ML Compelins

- AI/ML pipelines training data
- Building attack scripts or ML models
- Model repostiores
- Lack of dataset integrity validation
- APIS used of model access

Root Cause

- Publicly accessible training data
- Lack atte dataset integrity validation
- Untrusted thêrd-party models or libraries
- Weak supply-chain controls

Technical Impact



Poisoned ML models



Reduced model accuracy



Backbors embeded AI systems



Future exploluation made easier

Simple Preparation Flow Attacker



- ↓ Creates public data
- Injects into ML pipeline
- Compromised AI model

Pseuo-Code / Concept

Modify labels subttly
Upload as training dastasat

Prepared By: Intern – Sanika Jankar 2038

Attack flow:

MITRE ATLAS: Impact (AML.TA0040) - Attack Flow



Common Impact Techniques

- Dataset Poisoning
- Model Backdooring
- Malicious Pre-Trained Models
- C2 Infrastructure

MITRE ATLAS Mapping

Tactic: Impact (AML.TA0040)

Goal: Strategic preparation of assets to ensure the success a future AI attack

